

Qwen-Standard

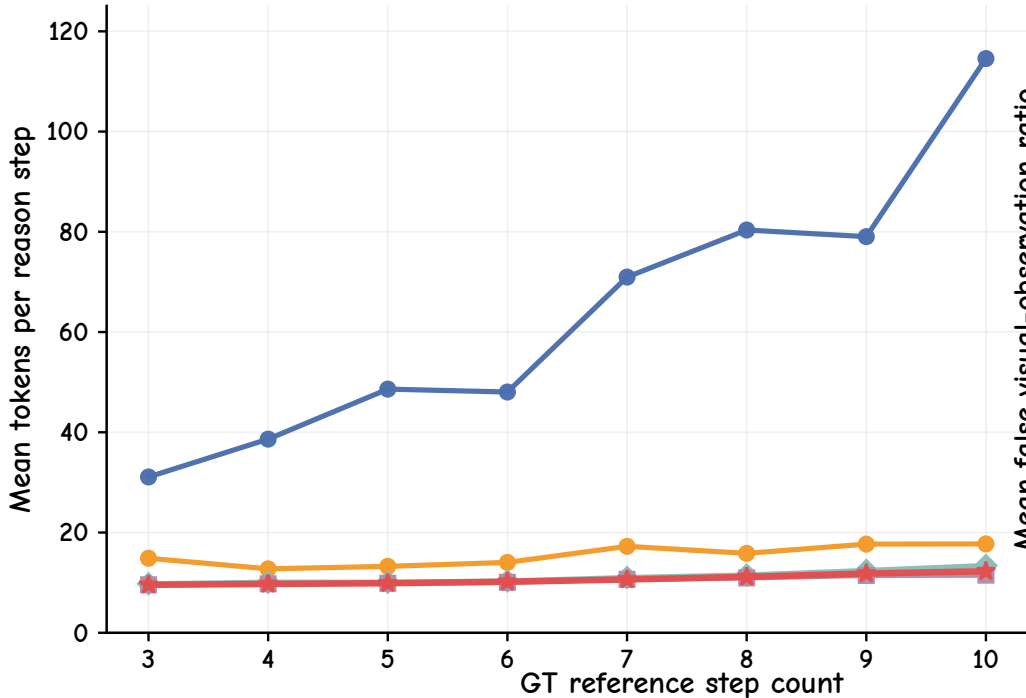
DRT-SFT

w/o process reward ($R = R_{\text{format}} + R_{\text{ans}}$)

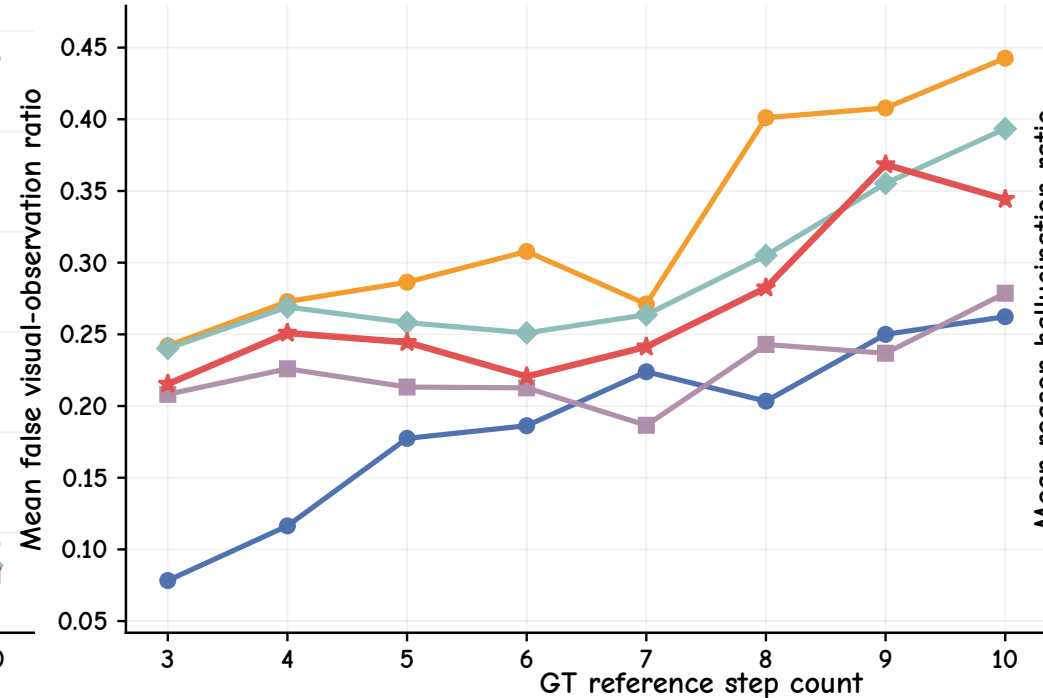
w/o step bonus ($R = R_{\text{format}} + R_{\text{main}}$)

DRT-RL

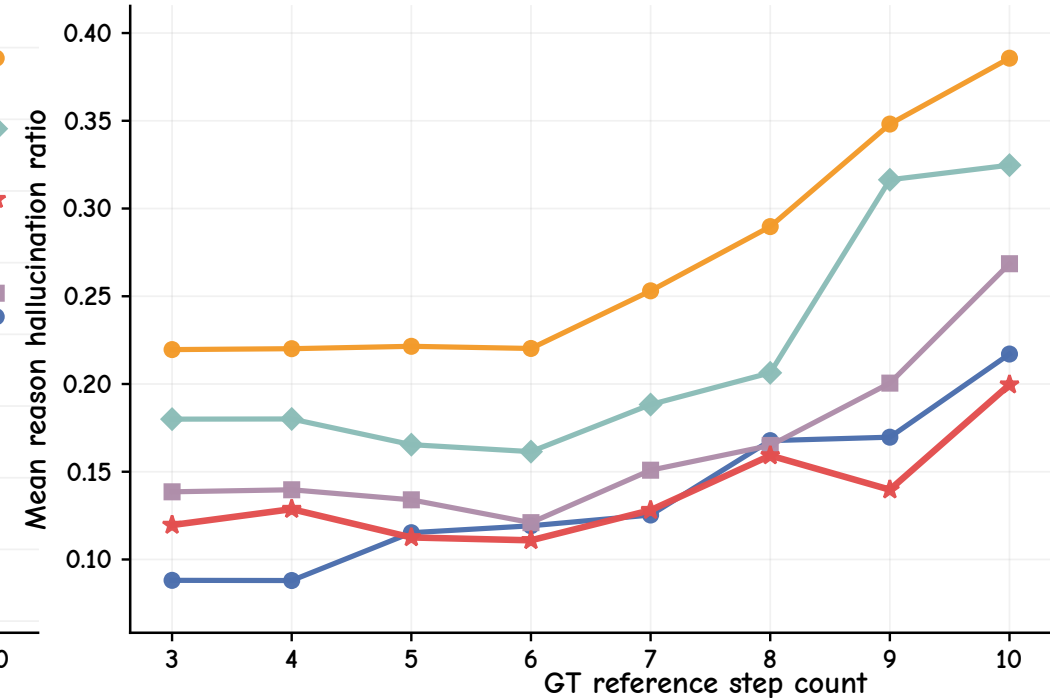
A. Token / Reason Step



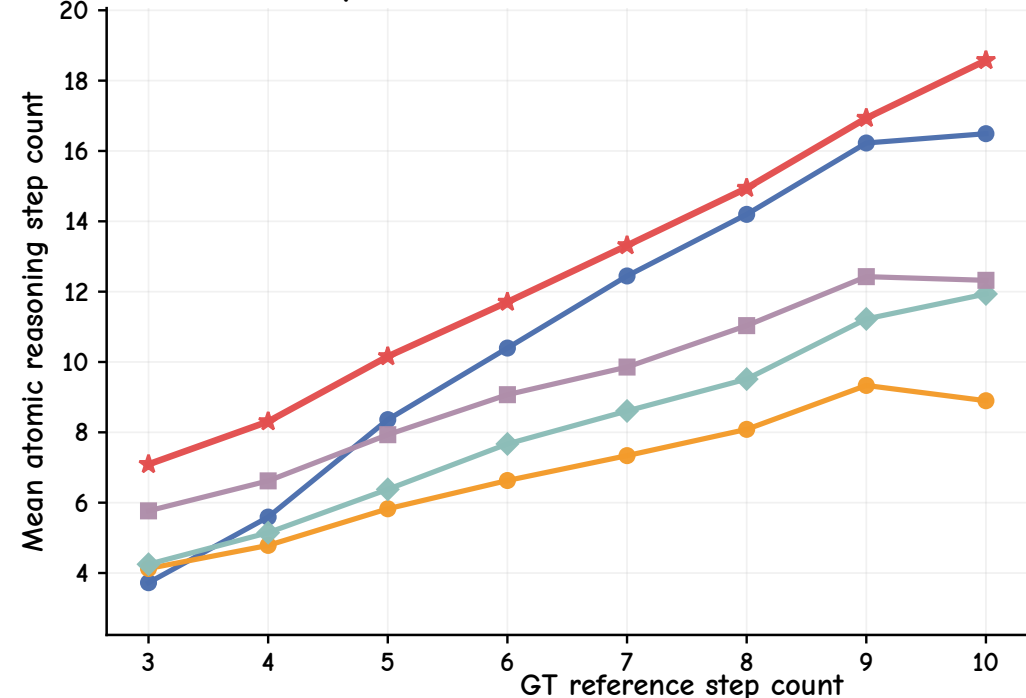
B. Visual Hallucination Ratio



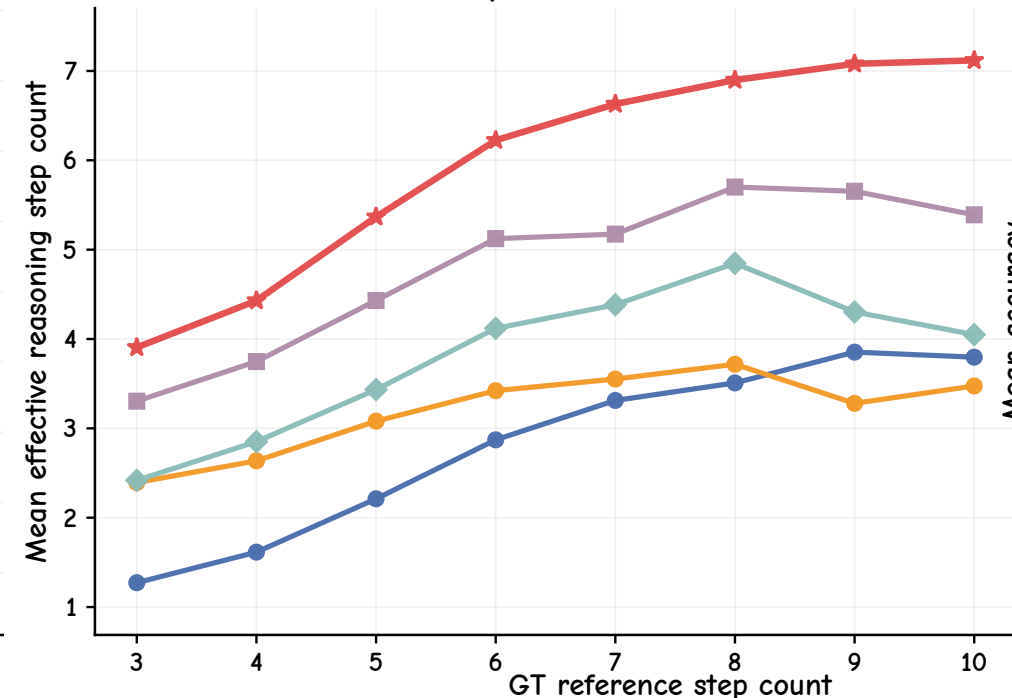
C. Reason Hallucination Ratio



D. Reason Step



E. Effective Reason Step



F. Accuracy

